

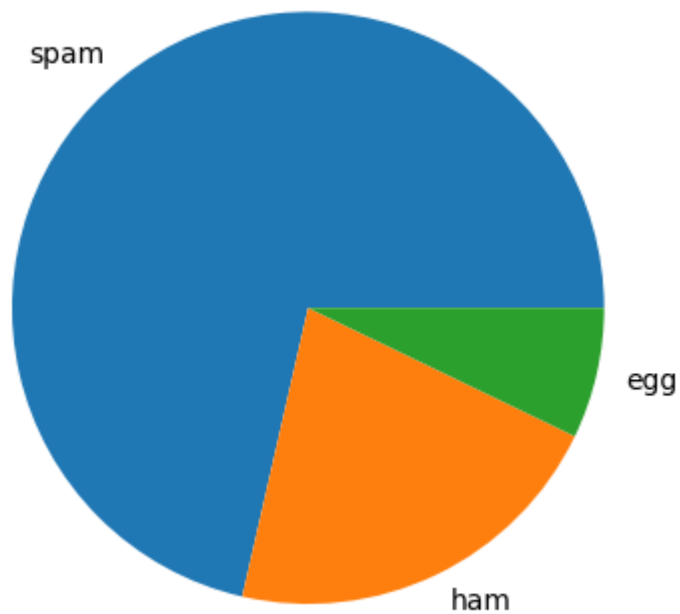
Lesson4 Matplotlib

201~208ページ

```
In [4]: import matplotlib.pyplot as plt
labels = ["spam", "ham", "egg"]
x = [10, 3, 1]

fig, ax = plt.subplots()

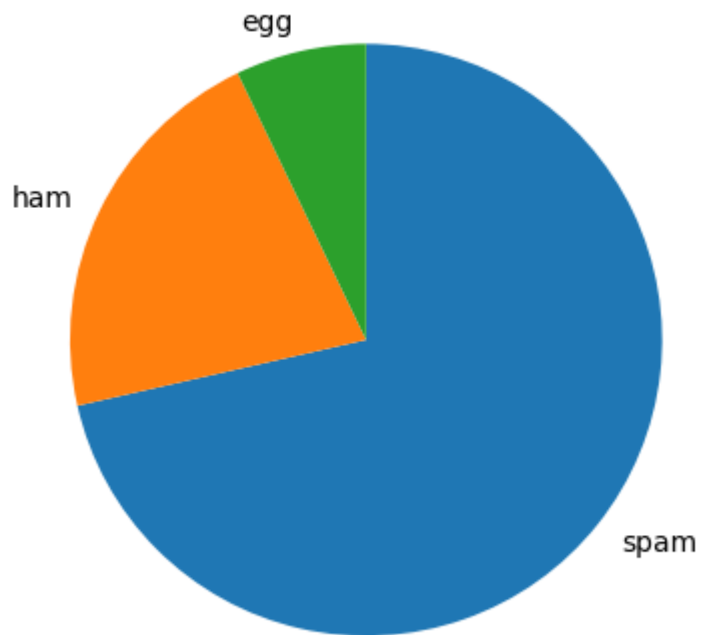
ax.pie(x, labels=labels)
plt.show()
```



```
In [5]: fig, ax = plt.subplots()

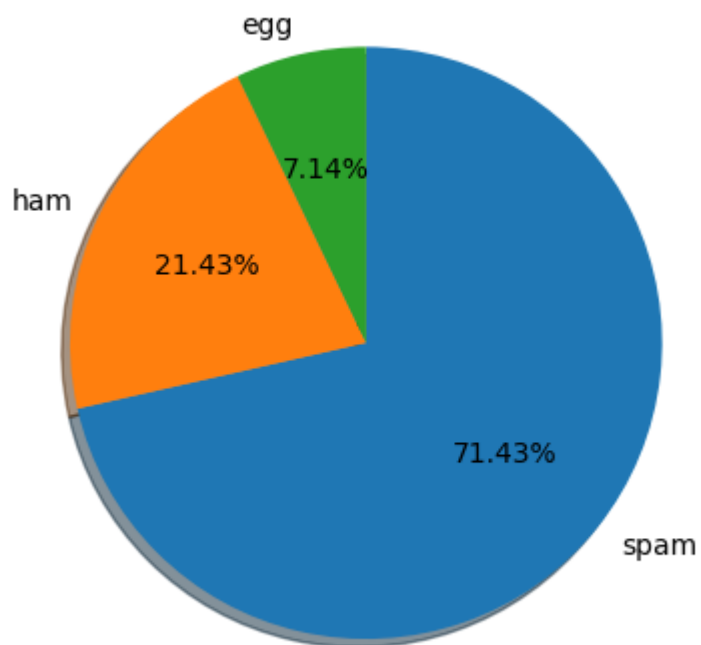
ax.pie(x, labels=labels, startangle=90, counterclock=False)

plt.show()
```



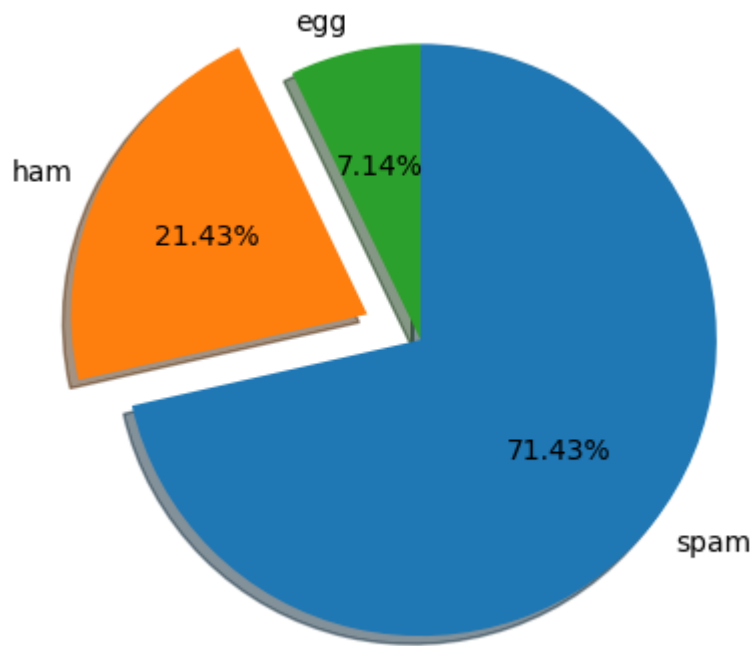
```
In [8]: fig, ax = plt.subplots()

ax.pie(x, labels=labels, startangle=90, counterclock=False, shadow=True, autopct=
plt.show()
```



```
In [9]: explode = [0,0.2,0]
fig, ax = plt.subplots()

ax.pie(x, labels=labels, startangle=90, counterclock=False, shadow=True, autopct=
plt.show()
```

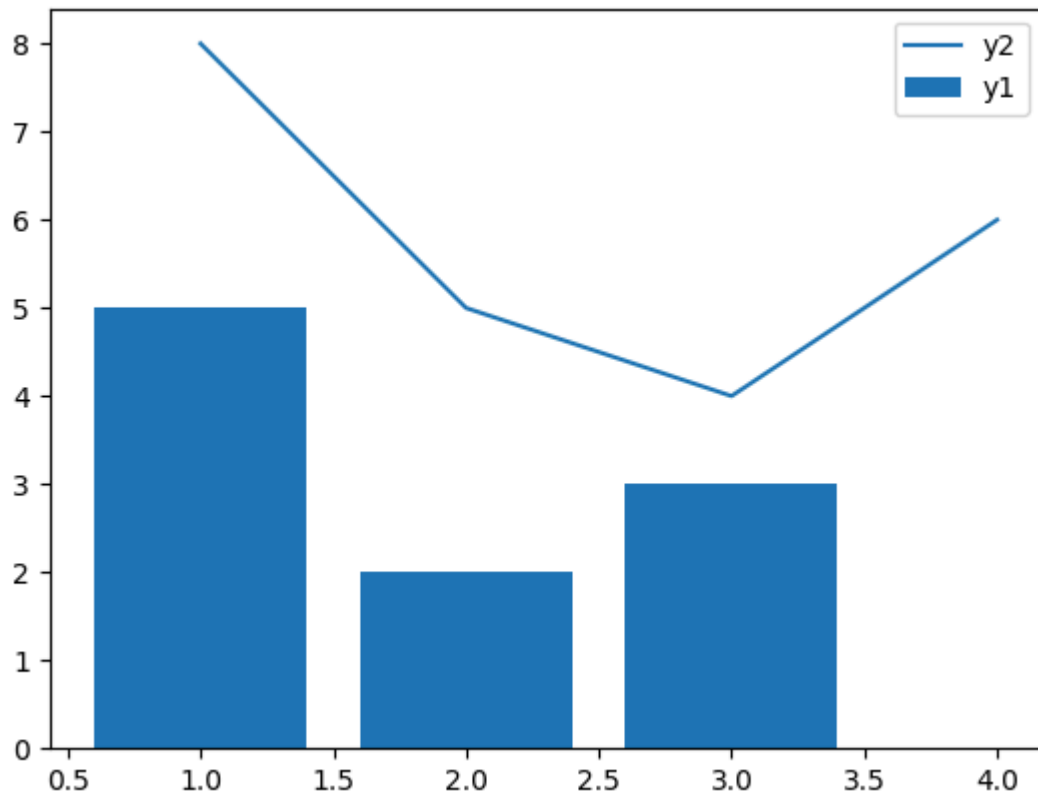


```
In [11]: fig, ax = plt.subplots()

x1 = [1,2,3]
y1 = [5,2,3]
x2 = [1,2,3,4]
y2 = [8,5,4,6]

ax.bar(x1,y1,label="y1")
ax.plot(x2,y2,label="y2")
ax.legend()

plt.show()
```



```
In [12]: import numpy as np

rng = np.random.default_rng(123)
x = rng.normal(size=1000)

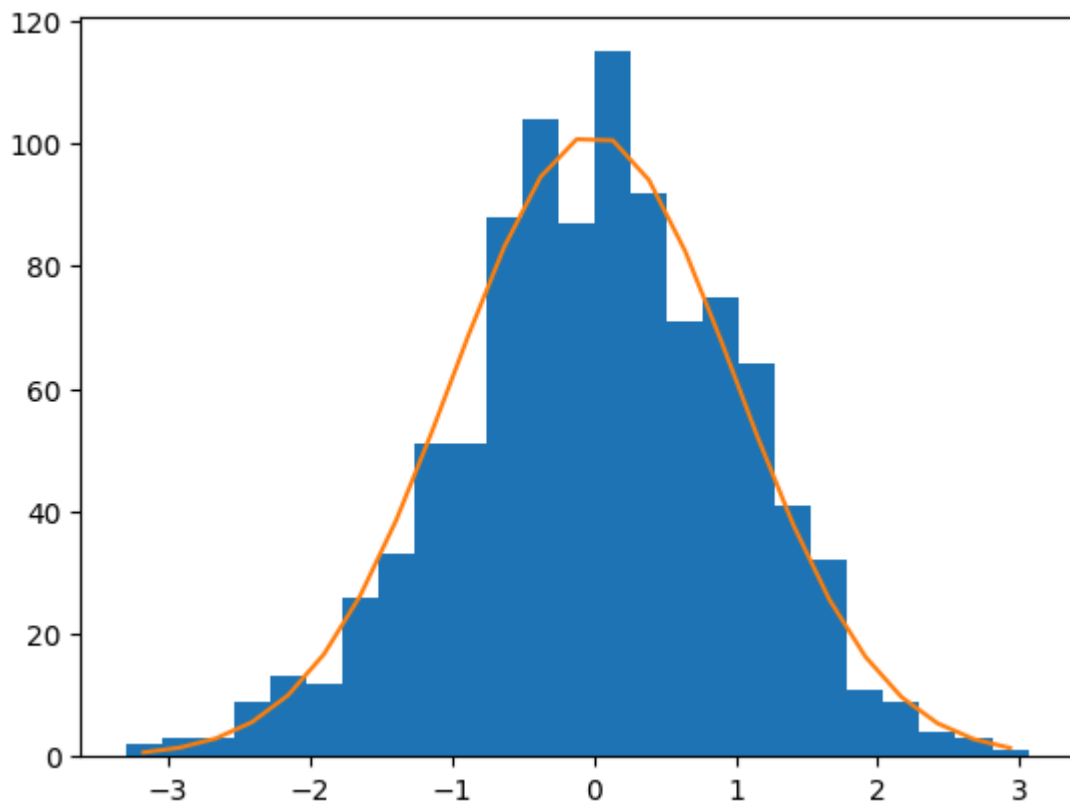
fig, ax = plt.subplots()

counts, edges, patches = ax.hist(x, bins=25)

x_fit = (edges[:-1] + edges[1:]) / 2

y = 1000 * np.diff(edges) * np.exp(-x_fit**2/2) / np.sqrt(2*np.pi)
ax.plot(x_fit, y)

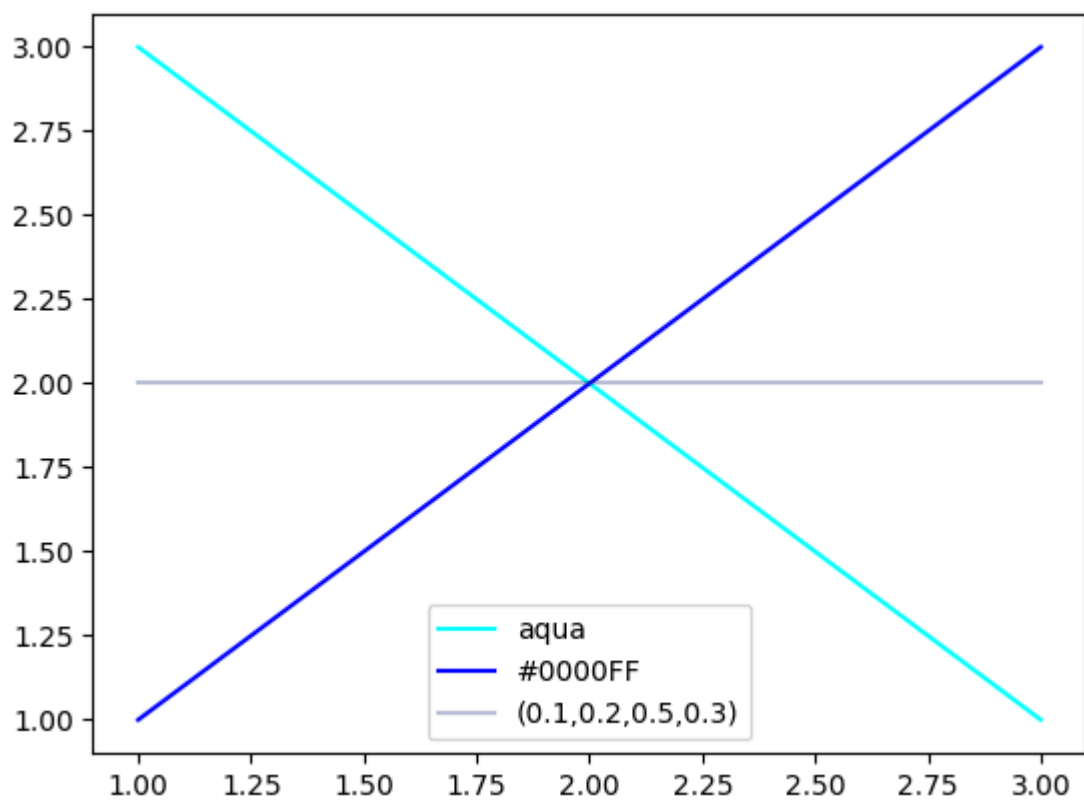
plt.show()
```



```
In [15]: fig, ax = plt.subplots()

ax.plot([1,3], [3,1], label="aqua", color="aqua")
ax.plot([1,3], [1,3], label="#0000FF", color="#0000FF")
ax.plot([1,3], [2,2], label="(0.1,0.2,0.5,0.3)", color=(0.1,0.2,0.5,0.3))
ax.legend()

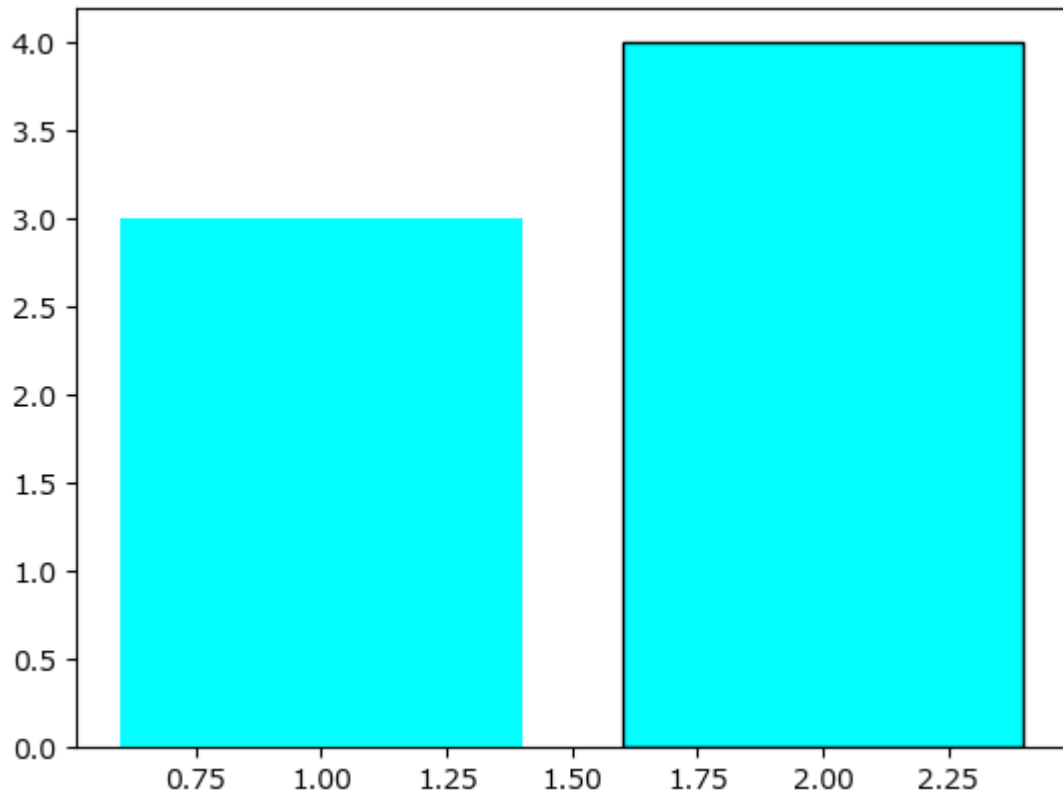
plt.show()
```



```
In [42]: fig, ax = plt.subplots()

ax.bar([1],[3], color="aqua")
ax.bar([2],[4], color="aqua", edgecolor="black")

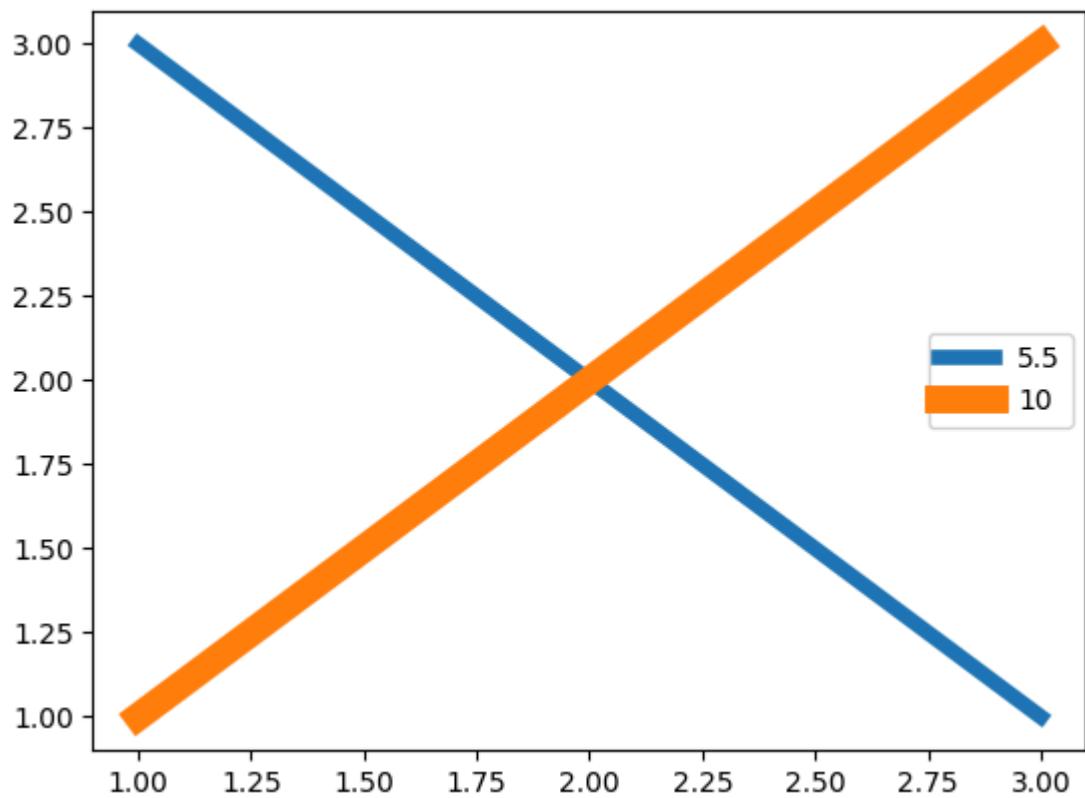
plt.show()
```



```
In [18]: fig, ax = plt.subplots()

ax.plot([1,3], [3,1], linewidth=5.5, label="5.5")
ax.plot([1,3], [1,3], linewidth=10, label="10")
ax.legend()

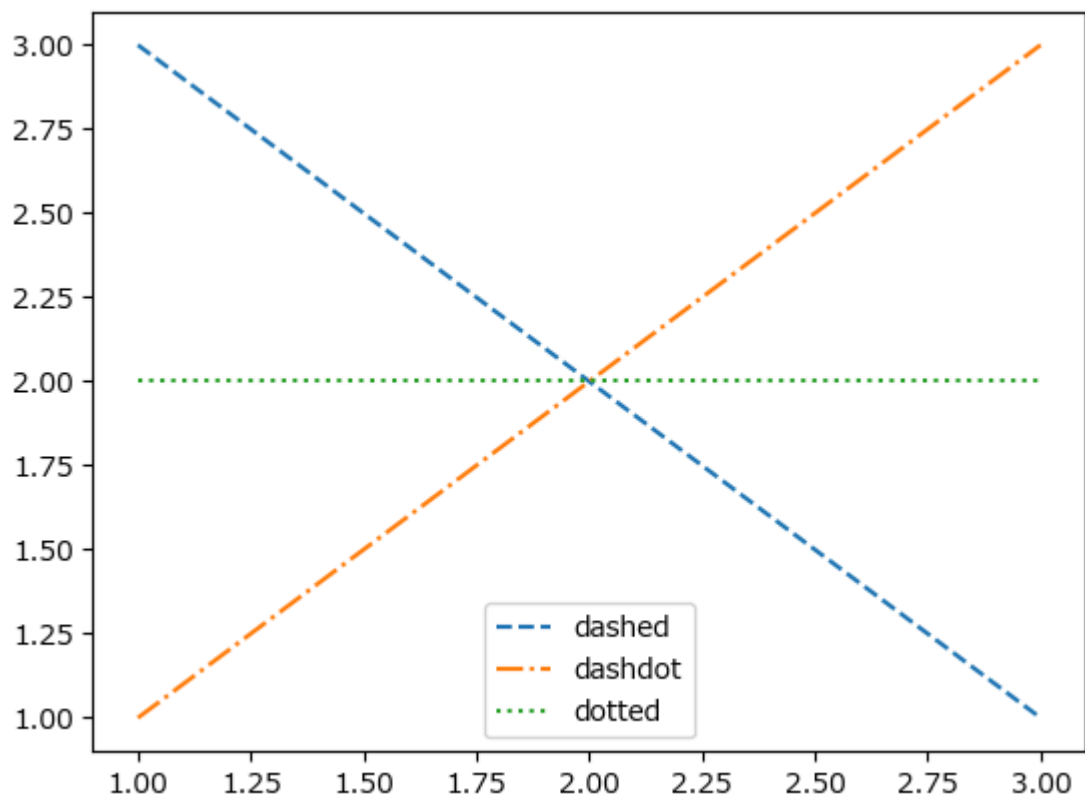
plt.show()
```



```
In [48]: fig, ax = plt.subplots()

ax.plot([1,3], [3,1], linestyle="--", label="dashed")
ax.plot([1,3], [1,3], linestyle="-. ", label="dashdot")
ax.plot([1,3], [2,2], linestyle=":", label="dotted")
ax.legend()

plt.show()
```



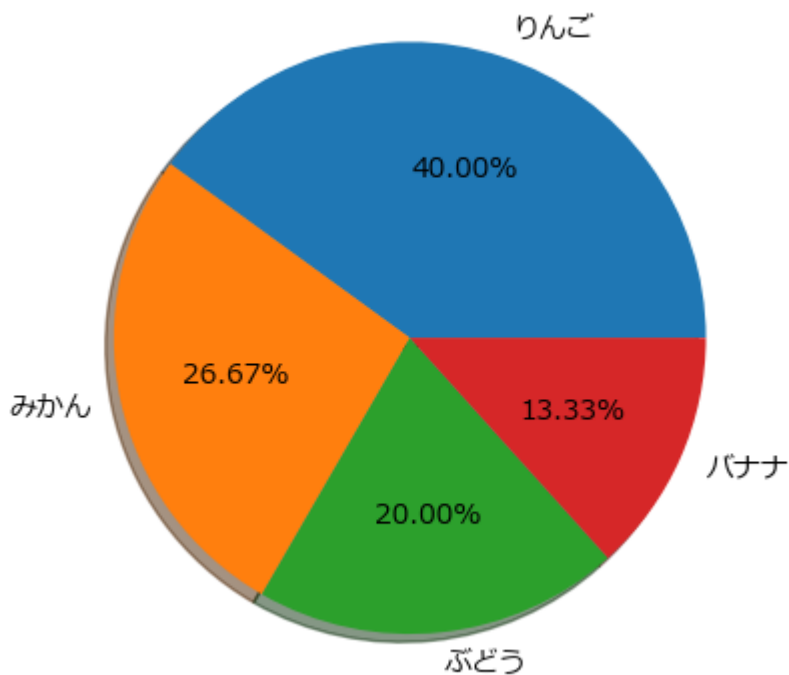
Matplotlib③ 演習問題

問1 円グラフの作成

```
In [25]: plt.rcParams["font.family"] = "Meiryo"
fig, ax = plt.subplots()
labels = ["りんご", "みかん", "ぶどう", "バナナ"]
x = [18, 12, 9, 6]

ax.pie(x, labels=labels, shadow=True, autopct="%1.2f%")

plt.show()
```



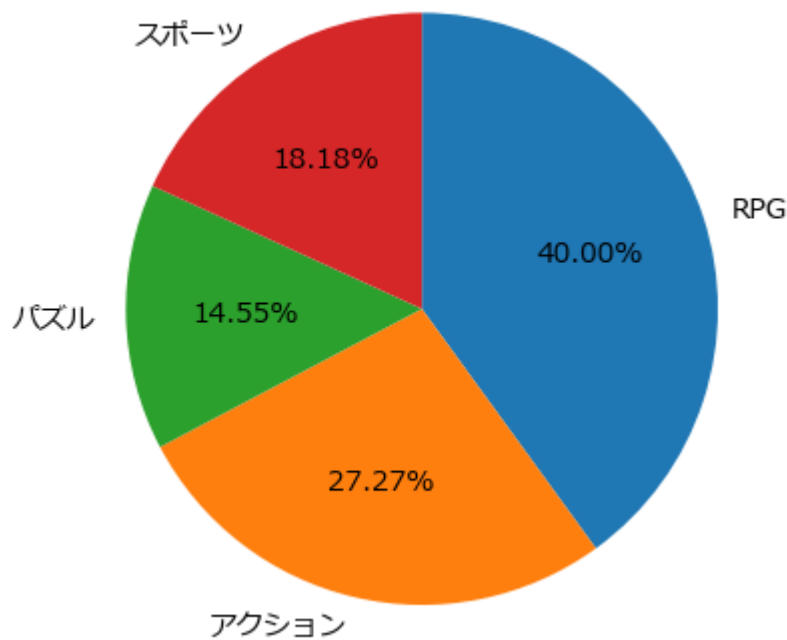
問2 人気投票

```
In [31]: labels=["RPG", "アクション", "パズル", "スポーツ"]
x=[22, 15, 8, 10]

fig, ax = plt.subplots()

ax.pie(x, labels=labels, startangle=90, counterclock=False, autopct="%1.2f%")

plt.show()
```

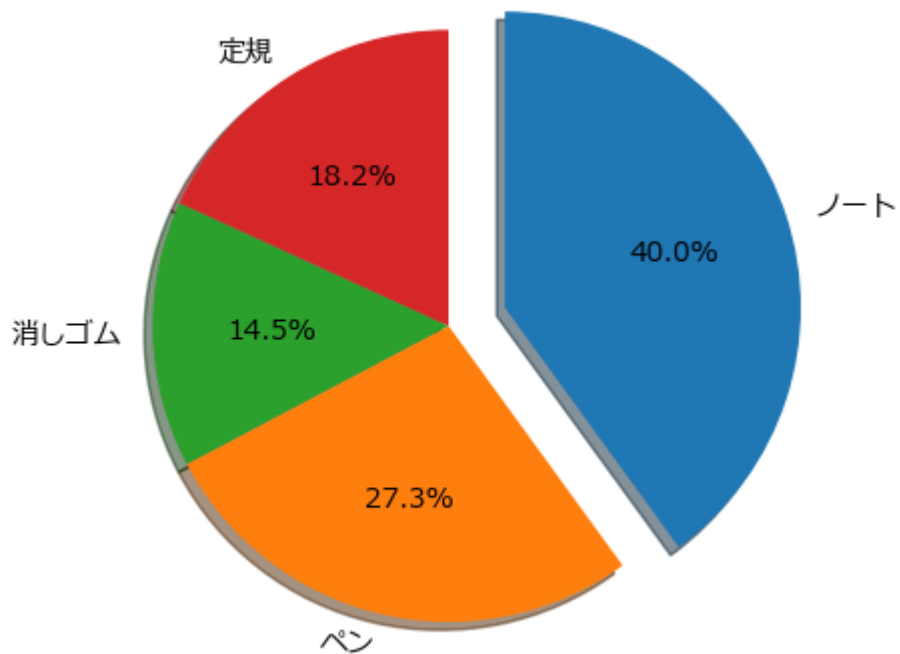
問3 売り上げ構成比

```
In [35]: labels=["ノート","ペン","消しゴム","定規"]
sales=[45,25,18,12]
explode = [0.2,0,0,0]

fig, ax = plt.subplots()

ax.pie(x, labels=labels, startangle=90, shadow=True, counterclock=False, autopct

plt.show()
```



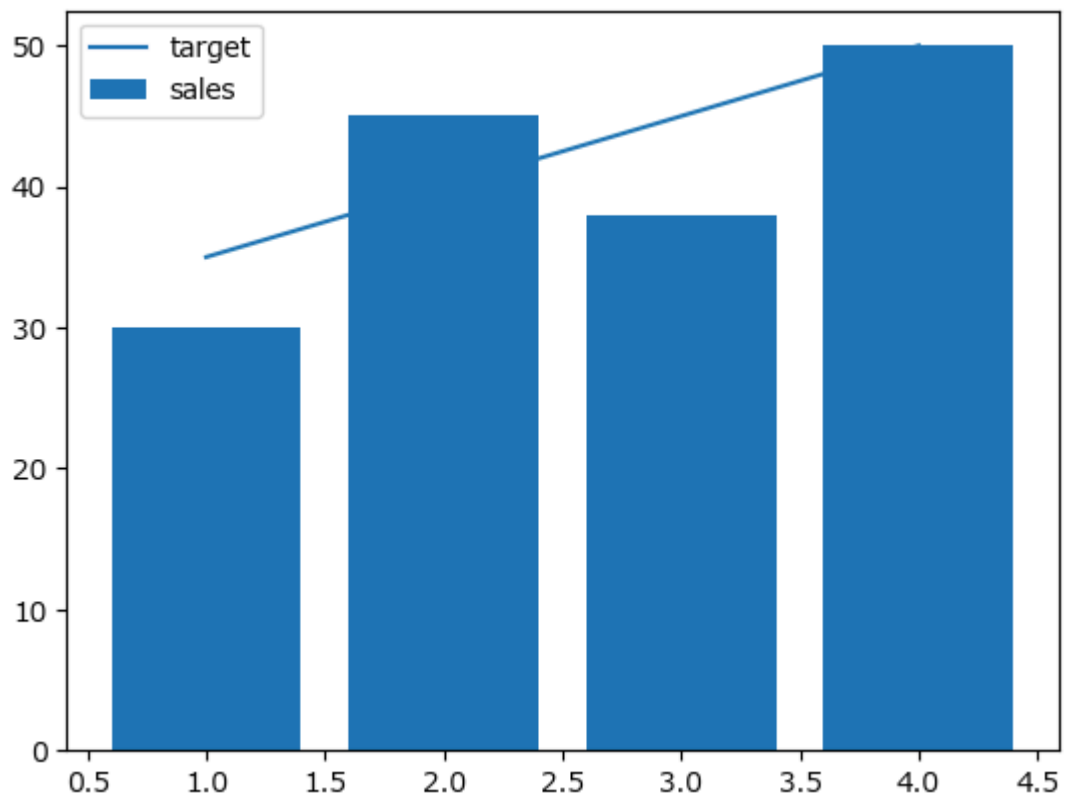
問4棒+折れ線

```
In [36]: month=[1,2,3,4]
sales=[30,45,38,50]
target=[35,40,45,50]

fig,ax = plt.subplots()

ax.bar(month,sales,label="sales")
ax.plot(month,target,label="target")
ax.legend()

plt.show()
```



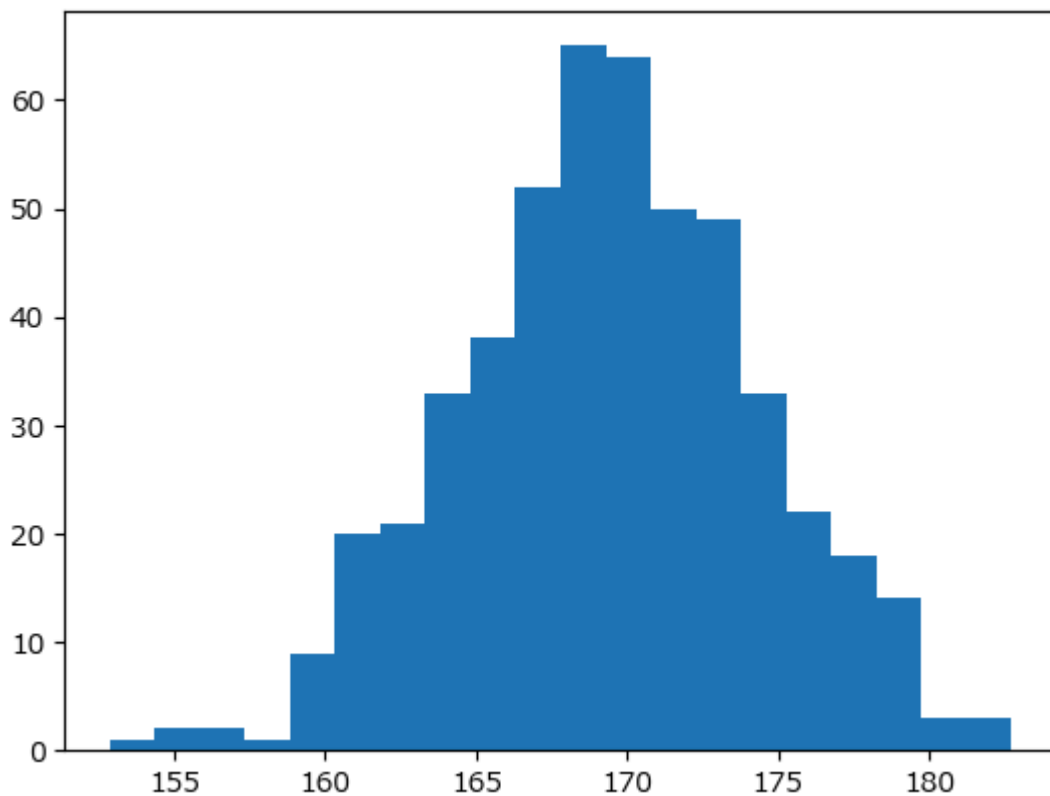
問5ヒストグラム

```
In [39]: rng=np.random.default_rng(10)
x=rng.normal(loc=170,scale=5,size=500)

fig,ax = plt.subplots()

counts, edges, patches = ax.hist(x, bins=20)

plt.show()
```

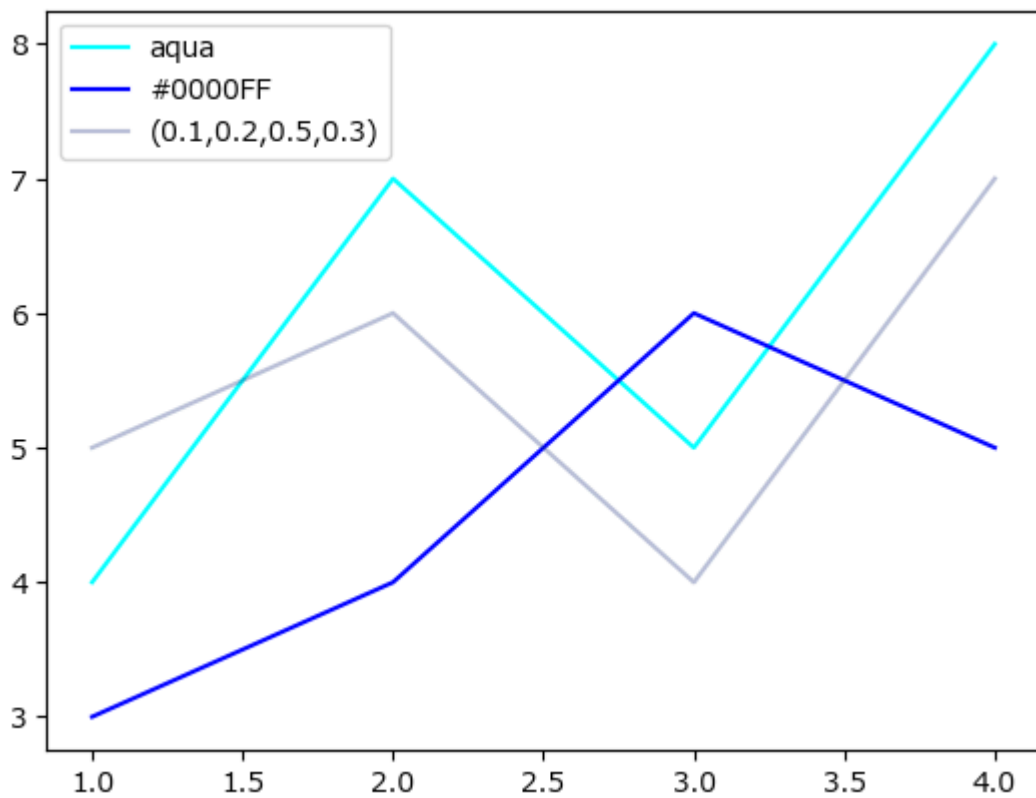


問 6 色指定

```
In [41]: fig, ax = plt.subplots()
x=[1,2,3,4]
y1=[4,7,5,8]
y2=[3,4,6,5]
y3=[5,6,4,7]

ax.plot(x, y1, label="aqua", color="aqua")
ax.plot(x, y2, label="#0000FF", color="#0000FF")
ax.plot(x, y3, label="(0.1,0.2,0.5,0.3)", color=(0.1,0.2,0.5,0.3))
ax.legend()

plt.show()
```



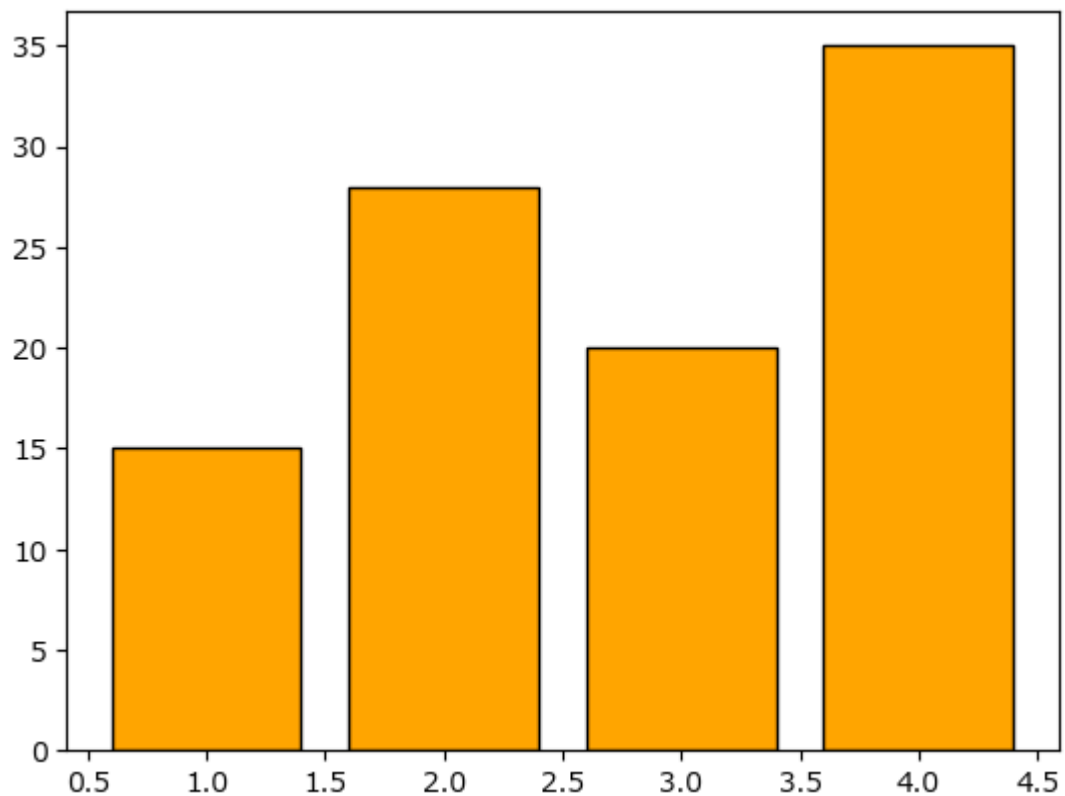
問 7 棒グラフ装飾

```
In [45]: item=[1,2,3,4]
sales=[15,28,20,35]

fig, ax = plt.subplots()

ax.bar(item, sales, color="orange", edgecolor="black")

plt.show()
```



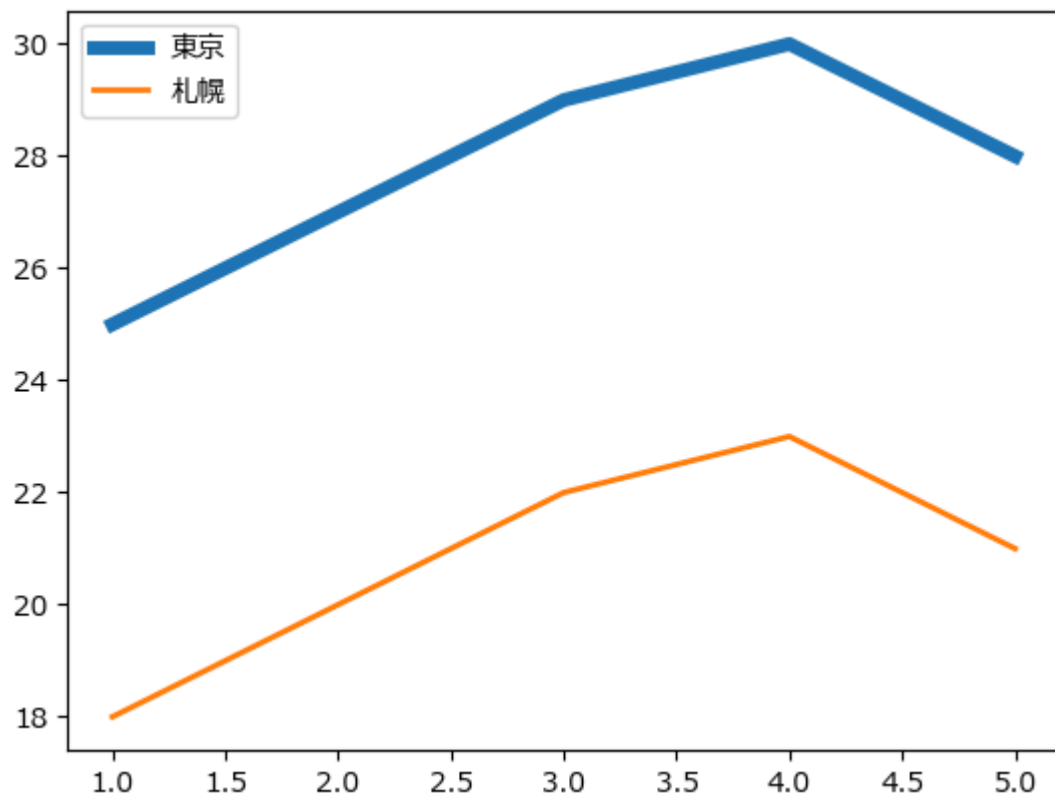
問 8 線幅

```
In [47]: day=[1,2,3,4,5]
         tokyo=[25,27,29,30,28]
         sapporo=[18,20,22,23,21]

         fig, ax = plt.subplots()

         ax.plot(day, tokyo, linewidth=5, label="東京")
         ax.plot(day, sapporo, linewidth=2, label="札幌")
         ax.legend()

         plt.show()
```



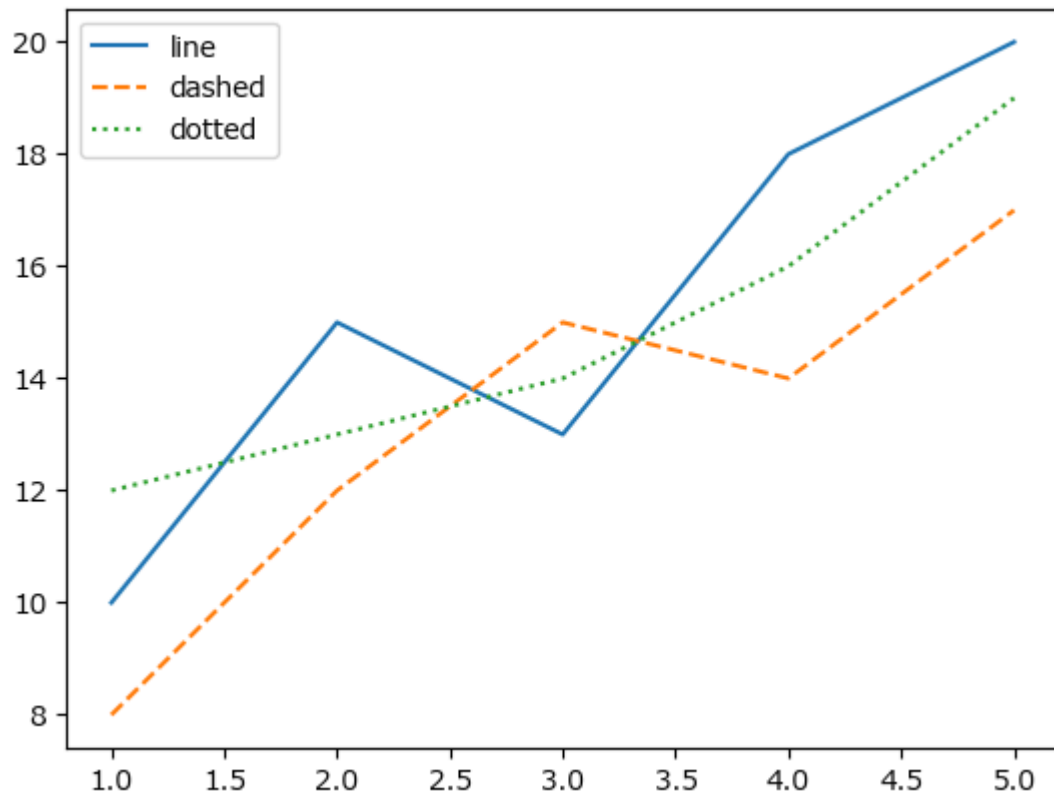
問9 線種

```
In [52]: x=[1,2,3,4,5]
a=[10,15,13,18,20]
b=[8,12,15,14,17]
c=[12,13,14,16,19]

fig, ax = plt.subplots()

ax.plot(x, a, linestyle="-", label="line")
ax.plot(x, b, linestyle="--", label="dashed")
ax.plot(x, c, linestyle=":", label="dotted")
ax.legend()

plt.show()
```



問 1 0 総合問題

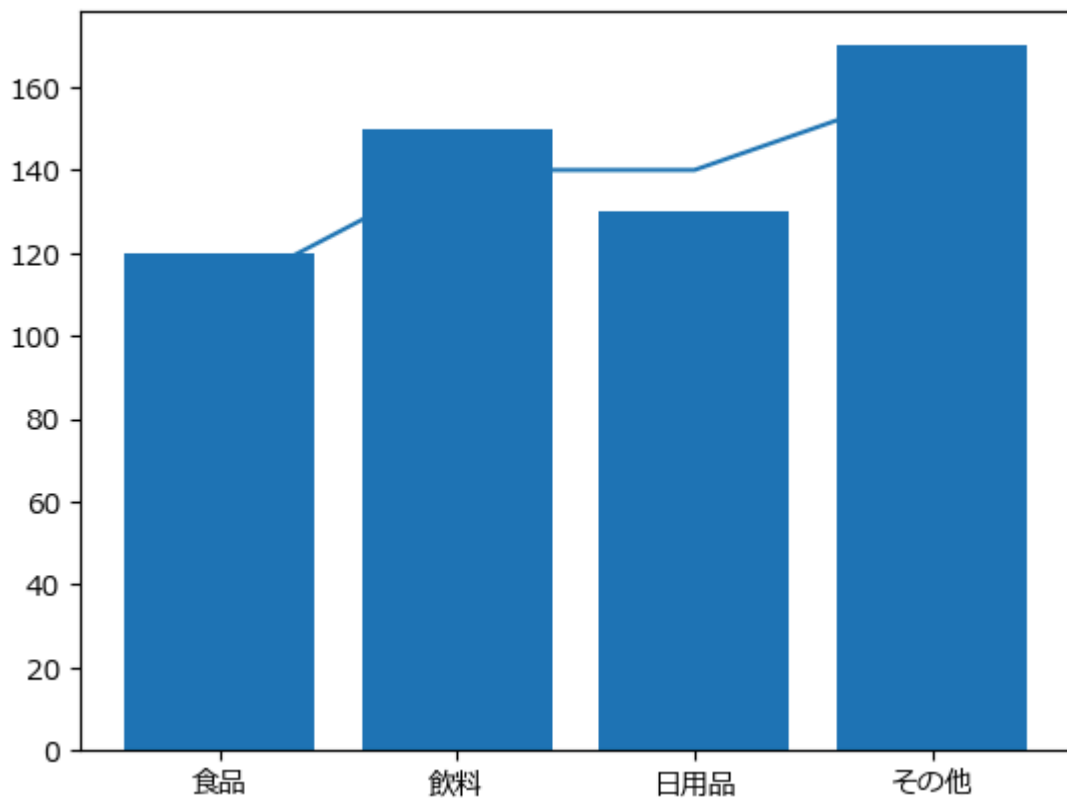
```
In [53]: months=[1,2,3,4]
sales=[120,150,130,170]
target=[110,140,140,160]
labels=["食品","飲料","日用品","その他"]
ratio=[40,25,20,15]
```

問題 1

```
In [59]: fig,ax = plt.subplots()

ax.bar(month,sales,tick_label=labels)
ax.plot(month,target)

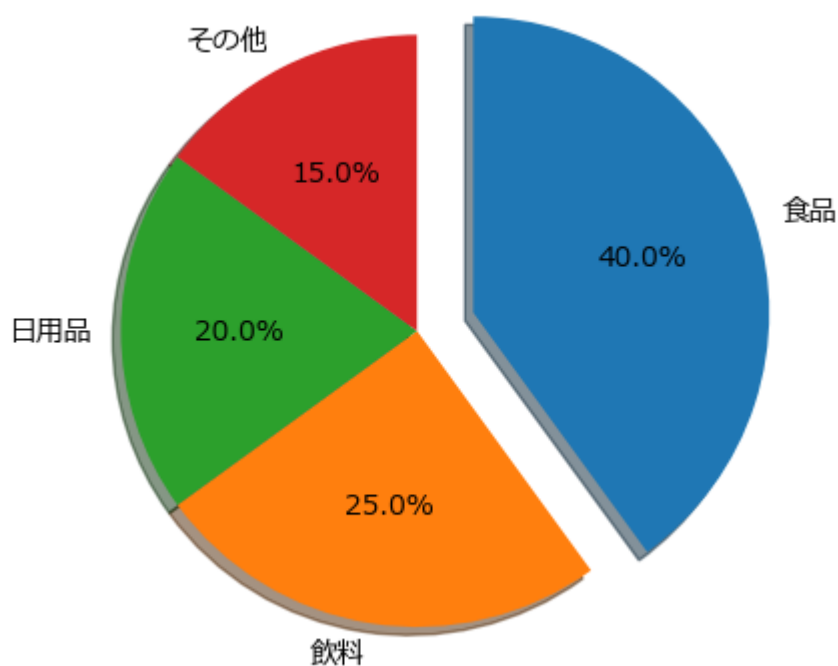
plt.show()
```

問題 2

```
In [61]: explode = [0.2,0,0,0]
fig, ax = plt.subplots()

ax.pie(ratio, labels=labels, startangle=90, shadow=True, counterclock=False, autopct='%1.1f%%')
plt.show()
```

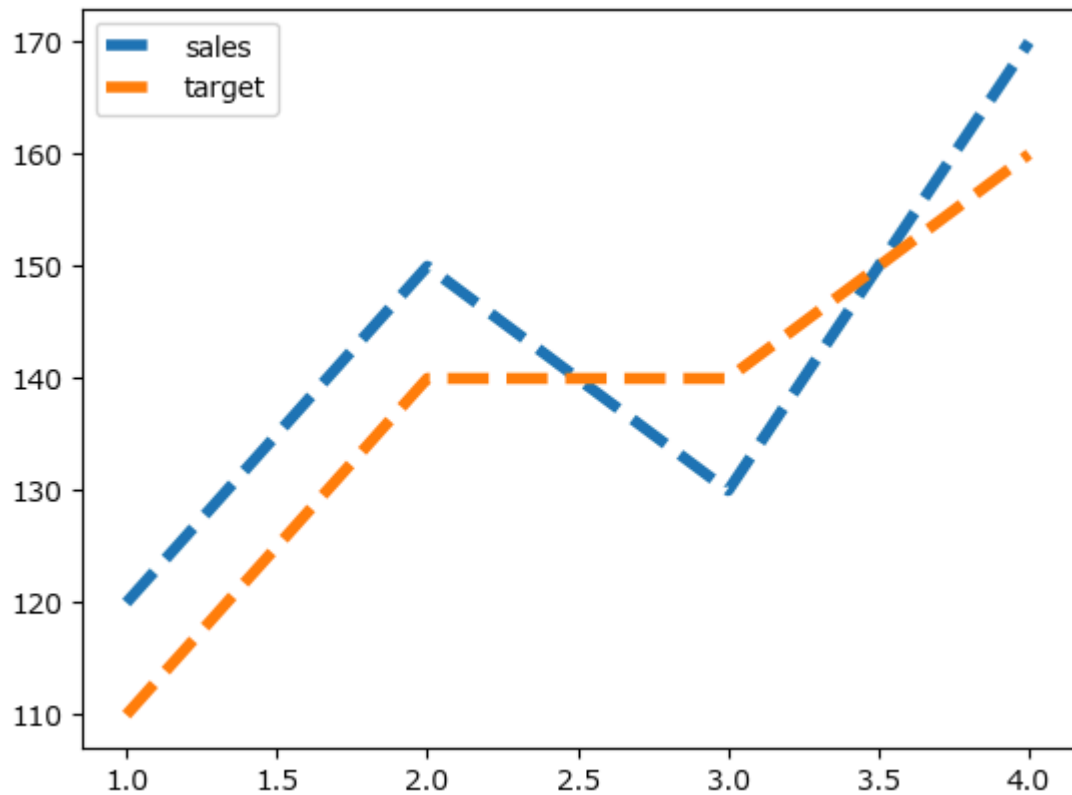


問題 3

```
In [63]: fig, ax = plt.subplots()

ax.plot(month, sales, linewidth = 4, linestyle="--", label="sales")
ax.plot(month, target, linewidth=4, linestyle="--", label="target")
ax.legend()

plt.show()
```



In []: